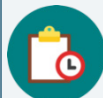


1.11 Perpendicular Bisectors

Lesson Introduction

 Benchmark	MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.		 Learning Targets	- I can construct a perpendicular bisector. - I can use rigid transformations to show that any triangle formed by drawing two segments from the endpoints of a segment to a point on its perpendicular bisector is isosceles. - I can use the properties of a perpendicular bisector to solve problems.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	- What key features define a perpendicular bisector? - How can a perpendicular bisector help in constructing an isosceles triangle? - If a perpendicular bisector exists in a triangle, which side lengths must be congruent?				
 Lesson Narrative	Students construct and examine perpendicular bisectors, determining that points on a perpendicular bisector are of equal distance to the endpoints and form an isosceles triangle. Students use the relationship between the line segment and the perpendicular bisector to solve problems.				
 Component Summary	1.11.1 Warm-Up (~5 min.)	Comparing the areas of three inscribed figures (<i>SE p. 69</i>)	1.11.4 Bring It Together (5-10 min.)	Relationships formed between a line segment and a perpendicular bisector (<i>SE p. 73</i>)	
	1.11.2 Exploration (10 min.)	Constructing and examining points on a perpendicular bisector (<i>SE p. 70</i>)	1.11.5 Guided Practice (5 min.)	Solving using a perpendicular bisector (<i>SE p. 73</i>)	
	1.11.3 Activity (10 min.)	Creating an isosceles triangle from a perpendicular bisector (<i>SE p. 72</i>)	1.11.6 Your Turn! (5 min.)	Practice with perpendicular bisectors (<i>SE p. 74</i>)	
	1.11.7 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 74</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Right Triangle - Isosceles Triangle <u>Additional Terms:</u> - Perpendicular Bisector		- Compass - Ruler - Patty Paper (Optional) String (Optional) Highlighters		Consider preparing sentence stems for question 6 in 1.11.2 Exploration.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not use a wide enough compass width to form intersecting arcs. - Students may change the width of their compass when measuring from each endpoint.	- If you are in an environment where students can go outside, consider scaling up these activities by replacing a compass with long string tied around a tree or post. - Consider using the GeoGebra applet for constructing a perpendicular bisector in 1.11.2 Exploration. - If time is an issue, then consider assigning 1.11.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talk When students begin 1.11.5 Guided Practice, a Number Talk would be a great way to get students discussing and working through these questions. This strategy should work well here because of the different ways for students to work through each equation.	MA.K12.MTR.2.1: Multiple Representations Students will build and demonstrate understanding by constructing a perpendicular bisector to represent equal distance to the endpoints of a line segment in 1.11.2 Exploration. Using manipulatives helps students make connections between concepts and representations.
 Differentiate Instruction	For the vocabulary in this section, have students identify the key words of the definition by highlighting or underlining each word that is critical to the definition. Students should then define or explain each of the key words. Once identified, have students read to each other the vocabulary word and the keys words. - Identify key words - Eliminate non-critical key words - Define each word - Share with partner	
Lesson Notes:		

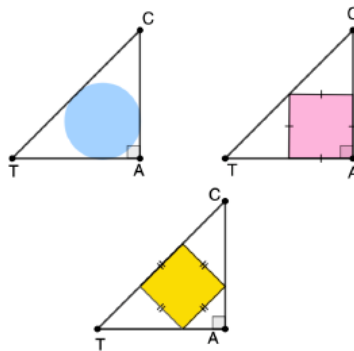
1.11.1 Warm-Up: Compare Areas

Component Overview: Students will justify which shaded region has the greatest area.



1.11.1 Warm-Up: Compare Areas

$\triangle CAT$ is an isosceles right triangle.



Students should be free to make their own determination and justification. Consider not requiring students to use precise language in their justification. If time, have students share their choice with a partner and to verbalize their justification.

- Which inscribed figure has the greatest area? Justify your choice.

Answers will vary. I think the triangle with a circle in it will have the largest area because the curved parts can cover more than a straight edge.

1.11.2 Exploration: Constructing Perpendicular Bisectors

Component Overview: Students construct the perpendicular bisector of a line segment. After measuring the distance from various points, students write a conjecture about the distance of any point on the perpendicular bisector to the endpoints of the line segment.



1.11.2 Exploration: Constructing Perpendicular Bisectors

1. Use the steps shown to construct the perpendicular bisector of $\overline{AB}$.

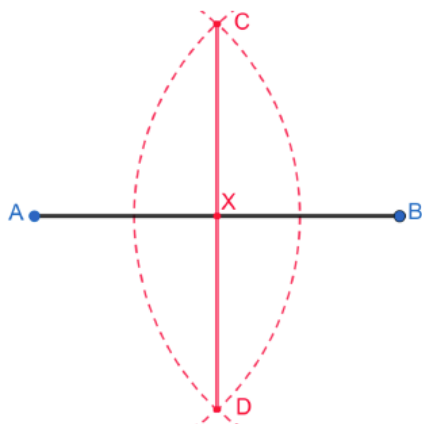
Step 1: Set the compass point at point A and open the compass to a width approximately two-thirds the length of $\overline{AB}$.

Step 2: Without changing the width of the compass, draw an arc above and below the line centered at point A .

Step 3: Repeat Steps 1 and 2 to create an arc centered at point B .

Step 4: Mark the two points of intersections of the arcs as C and D . Draw a line creating $\overline{CD}$ that intersects $\overline{AB}$.

Step 5: Label the point of intersection for $\overline{CD}$ and $\overline{AB}$ as point X .



Consider using the GeoGebra applet for 1.11.2 Exploration.



1.11.2 Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize in 1.11.4 Bring It Together. Students may revise their thinking throughout the exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 1.11.4 Bring It Together.



Students typically struggle placing arcs that intersect. As you circulate, ensure that students are using compass widths that are greater than half the length of the line segment. Students may need to be reminded to keep their compass setting the same when locating point C and also the same when locating point D .

1.11.2 Exploration: Constructing Perpendicular Bisectors (continued)

2. Use a protractor and ruler to find each of the measurements.

Answers will vary depending on students' compass placement. The measures of segments AX and BX are half of the measurement found for segment AB . $CB = CA$ and $DB = DA$.

Angle	Measurement	Segment	Measurement
$\angle AXD$	90°	$\overline{AB}$	6.2 cm
		$\overline{AX}$	3.1 cm
$\angle AXC$	90°	$\overline{BX}$	Answers will vary.
		$\overline{CB}$	Answers will vary.
$\angle BXD$	90°	$\overline{DB}$	Answers will vary.
		$\overline{CA}$	Answers will vary.
$\angle BXC$	90°	$\overline{DA}$	Answers will vary.



Students can use any unit to measure the line segments but must use the same unit of measure for all of them.

3. Complete the statements.

The angle formed by the line of intersection of $\overline{AB}$ and $\overline{CD}$ is

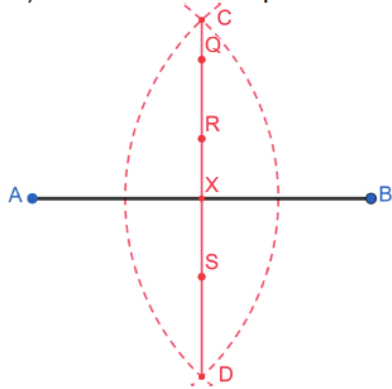
- ☐ 85°
☒ 90°
☐ 105°

The perpendicular bisector $\overline{CD}$ divides

- ☒ 2
☐ 3
☐ 4

$\overline{AB}$ into _____ congruent parts.

4. On the perpendicular bisector $\overline{CD}$, add three points labeled Q , R , and S in different places.



1.11.2 Exploration: Constructing Perpendicular Bisectors (continued)

5. Using a ruler, measure the length of each line segment.

Answers will vary depending on the placement of the points along $\overline{CD}$.

Segment	Length
$\overline{QA}$	
$\overline{QB}$	
$\overline{RA}$	
$\overline{RB}$	
$\overline{SA}$	
$\overline{SB}$	

6. Write a conjecture about the endpoints of a line segment and any point on its perpendicular bisector.

Answers will vary. The distance from an end point to any point on the perpendicular bisector will be the same length as the other end point of the same line and that point on the perpendicular bisector.

7. Share your conjecture with a partner. Summarize your partner's conjecture and have them initial that your summary is correct.

Answers will vary.

Partner's Conjecture	Partner's Initials



Creating a conjecture can be intimidating for students. Consider providing students with sentence stems to help scaffold their thinking.

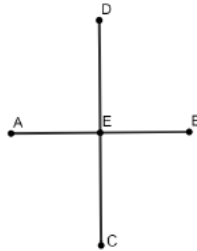
1.11.3 Activity: Perpendicular Bisectors and Triangles

Component Overview: Students use patty paper and transformations to determine the definition of an isosceles triangle.

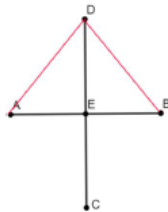


1.11.3 Activity: Perpendicular Bisectors and Triangles

- Copy $\overline{AB}$ and its perpendicular bisector $\overline{CD}$ on patty paper.



- Draw $\overline{AD}$ and $\overline{BD}$.

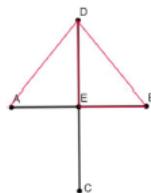


- Make a conjecture about $\triangle ADE$ and $\triangle BDE$.
Answers will vary. Both triangles are right triangles. The lines drawn to make the triangles are both the same lengths.



A **right triangle** is a triangle containing an interior right angle.

- Reflect $\triangle ADE$ over $\overline{CD}$.



1.11.3 Activity supports vocabulary terms that students should already be familiar with while also foreshadowing SAS congruence of triangles. The congruence of the triangles is shown using transformations which spirals the content previously learned in Unit 1.

1.11.3 Activity: Perpendicular Bisectors and Triangles (continued)

5. Discuss with your partner what the reflection shows is true about $\triangle ADE$ and $\triangle BDE$. Summarize your discussion.
The triangles are congruent.



An **isosceles triangle** contains at least two equal length sides and two equal interior angle measures.

6. Complete the statements.

In $\triangle ABD$, $\angle DAB$ is congruent to ☐ $\angle ADB$
☐ $\angle DBA$ because a

reflection is a rigid motion that preserves angle measure.

Since $m\angle DAB$ is equal to ☐ $m\angle ADB$,
☐ $m\angle DBA$, and ☐ $AD = AB$,
☐ $AD = BD$,
☐ $AE = BE$,
☐ None of these

by definition $\triangle ABD$ is a(n) ☐ scalene
☐ **isosceles**
☐ equilateral triangle.



Consider asking one of the partner pairs to read the completed statements to the class as a way to verify their choices. This will provide an auditory entry-point for learners from listening.

1.11.4 Bring It Together: Perpendicular Bisectors and Triangles

Component Overview: Students develop and test statements about points on a perpendicular bisector. They then use a ruler to confirm the definitions.



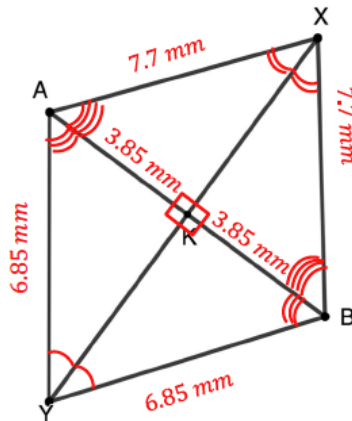
1.11.4 Bring It Together: Perpendicular Bisectors and Triangles



If a point is on the perpendicular bisector of a line segment, then it is equidistant from the endpoints of the line segment.

In the diagram, $\overline{XY}$ is the perpendicular bisector of $\overline{AB}$, $XA=7.7$ millimeters (mm), $YB=6.85$ mm, and $AB=7.7$ mm. The figure is not drawn to scale.

1. Label all segment measurements from the given information on the image. Then, label all other segments based on the relationship of a line segment and its perpendicular bisector.



2. Given $\overline{XY}$ is the perpendicular bisector of $\overline{AB}$, label all of the congruent angles.



The triangle formed by a line segment and two segments drawn from the endpoints of the segment to a point on the perpendicular bisector of the segment is an isosceles triangle.



Consider debriefing any questions that students had in 1.11.2 Exploration and 1.11.3 Activity. Also, consider highlighting student voices to share rich discussions or comments overhead while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.



After students have labeled the segments from the given information, consider asking the following questions:

- Is the length of $\overline{YK}$ equal to the length of $\overline{XK}$? How do you know?
- How could you determine the lengths of $\overline{YK}$ and $\overline{XK}$?

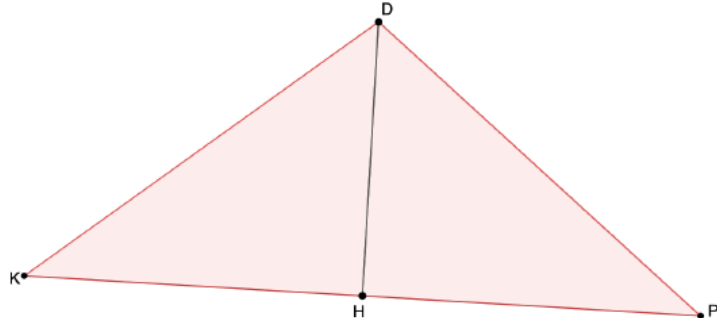
1.11.5 Guided Practice: Perpendicular Bisectors

Component Overview: Students use a perpendicular bisector to solve for an unknown variable and a segment length.



1.11.5 Guided Practice: Perpendicular Bisectors

1. $\overline{DH}$ is the perpendicular bisector of $\overline{KP}$, where $KD = -x - 1$ and $PD = -\frac{x}{2} + 19$.



- What is the value of x ?
 $x = -40$
- What is the length of $\overline{KD}$?
 $KD = 39$



Allow informal observation throughout the previous two components to dictate how “guided” 1.11.5 Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

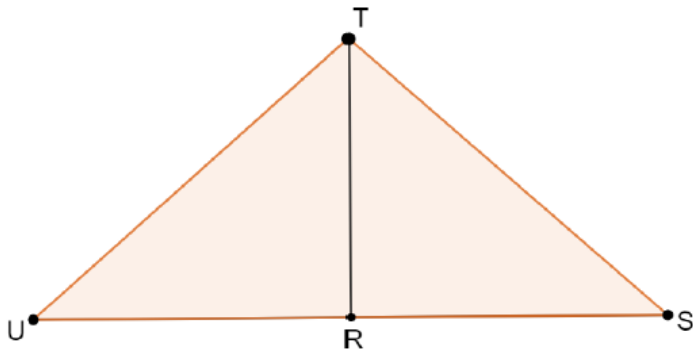
1.11.6 Your Turn!

Component Overview: Students use the perpendicular bisector to solve for an unknown variable. Then, a construction with an error is shown, where students describe the error.



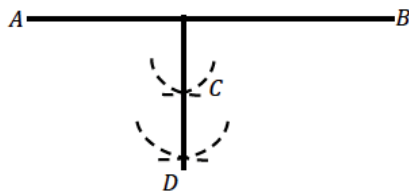
1.11.6 Your Turn!

- $\overline{RT}$ is the perpendicular bisector of $\overline{US}$. $UT = 8y - 12$ and $ST = 3y + 23$. Find the value of y .
 $y = 7$



- Describe the error that was made in the construction of a perpendicular bisector.

Answers will vary. When drawing arcs to locate point C, the compass width was not kept the same when drawing the arcs from point A and point B. The same error was made when locating point D.



If students struggle with finding the error, have the students use a compass to try and replicate the arcs. They will find that the compass width did not remain the same when locating point C or when locating point D. Question students on why it is important that the compass width stays the same. (Answer: the compass width maintains equal distance from both endpoints of the line segment).

1.11.7 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



1.11.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.
Answers will vary.



The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.